

Linear First-Order Equations & the Integrating Factor

ANSWER KEY



Recognizing linear form

- 1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.
Already in standard form: $P(x) = 2x$, $Q(x) = x$.
 - 2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.
Yes. Standard form: $y' - \frac{1}{x}y = x^2$, so $P(x) = -\frac{1}{x}$, $Q(x) = x^2$.
 - 3 Is $y' = y^2 + x$ a linear first-order equation? Justify your answer.
No. A linear equation cannot contain y^2 or any other nonlinear function of y ; the dependent variable and its derivatives must appear only to the first power and never multiplied together.
 - 4 Write $xy' + 3y = \sin x$ ($x > 0$) in standard form.
Divide by x : $y' + \frac{3}{x}y = \frac{\sin x}{x}$, so $P(x) = \frac{3}{x}$, $Q(x) = \frac{\sin x}{x}$.
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Finding the integrating factor

- 5 Find the integrating factor for $y' + 3y = e^x$.
 $\mu(x) = e^{\int 3 dx} = e^{3x}$.
 - 6 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.
 $\mu(x) = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2}$.
 - 7 Find the integrating factor for $y' + \frac{1}{x}y = 3x$, $x > 0$.
 $\mu(x) = e^{\int (1/x) dx} = e^{\ln x} = x$.
 - 8 Find the integrating factor for $y' + 2xy = 4x$.
 $\mu(x) = e^{\int 2x dx} = e^{x^2}$.
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Solving with the integrating factor

- 9 Solve $y' + 3y = e^x$.
With $\mu = e^{3x}$: $(e^{3x}y)' = e^{4x}$, so $e^{3x}y = \frac{1}{4}e^{4x} + C$, giving $y = \frac{1}{4}e^x + Ce^{-3x}$.
- 10 Solve $y' - \frac{2}{x}y = x$, $x > 0$.
With $\mu = x^{-2}$: $(x^{-2}y)' = x^{-1}$, so $x^{-2}y = \ln x + C$, giving $y = x^2 \ln x + Cx^2$.
- 11 Solve $y' + \frac{1}{x}y = 3x$, $x > 0$.
With $\mu = x$: $(xy)' = 3x^2$, so $xy = x^3 + C$, giving $y = x^2 + \frac{C}{x}$.

- 12 Solve $y' + 2xy = 4x$.

With $\mu = e^{x^2}$: $(e^{x^2}y)' = 4xe^{x^2}$. Since $\int 4xe^{x^2} dx = 2e^{x^2} + C$, $e^{x^2}y = 2e^{x^2} + C$, giving $y = 2 + Ce^{-x^2}$.

Initial value problems

- 13 Solve $y' + y = 2$, $y(0) = 0$.

With $\mu = e^x$: $(e^xy)' = 2e^x$, so $e^xy = 2e^x + C$, giving $y = 2 + Ce^{-x}$. Since $y(0) = 0$, $C = -2$:
 $y = 2 - 2e^{-x}$.

- 14 Solve $y' + 2y = 4x$, $y(0) = 1$.

With $\mu = e^{2x}$: $(e^{2x}y)' = 4xe^{2x}$. By parts, $\int 4xe^{2x} dx = 2xe^{2x} - e^{2x} + C$. So $y = 2x - 1 + Ce^{-2x}$. Since $y(0) = 1$, $C = 2$: $y = 2x - 1 + 2e^{-2x}$.

- 15 Solve $y' - y = e^{2x}$, $y(0) = 3$.

With $\mu = e^{-x}$: $(e^{-x}y)' = e^x$, so $e^{-x}y = e^x + C$, giving $y = e^{2x} + Ce^x$. Since $y(0) = 3$: $1 + C = 3 \Rightarrow C = 2$.
So $y = e^{2x} + 2e^x$.

- 16 Solve $y' + \frac{2}{x}y = \frac{\cos x}{x^2}$, $x > 0$, $y(\pi) = 0$.

With $\mu = x^2$: $(x^2y)' = \cos x$, so $x^2y = \sin x + C$. Since $y(\pi) = 0$: $\pi^2(0) = \sin \pi + C \Rightarrow C = 0$. So $y = \frac{\sin x}{x^2}$.

Application: a filling tank

- 17 A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t , the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.

Standard form $Q' + 0.03Q = 6$. With $\mu = e^{0.03t}$: $(e^{0.03t}Q)' = 6e^{0.03t}$, so $e^{0.03t}Q = 200e^{0.03t} + C$, giving $Q = 200 + Ce^{-0.03t}$. Since $Q(0) = 0$, $C = -200$: $Q(t) = 200(1 - e^{-0.03t})$.

- 18 For the tank above, what is the long-run (as $t \rightarrow \infty$) salt content, and why does that match the tank's incoming brine concentration times its volume?

As $t \rightarrow \infty$, $e^{-0.03t} \rightarrow 0$, so $Q(t) \rightarrow 200$ kg. This matches $2 \text{ kg/L} \times 100 \text{ L} = 200 \text{ kg}$ — once the transient dies out, the tank equilibrates to the same concentration as the inflow.

Application: an RL electrical circuit

- 19 An RL circuit with resistance R , inductance L , and constant EMF E_0 satisfies $L \frac{dl}{dt} + Rl = E_0$. Rewrite this in standard linear form and identify its integrating factor.

Standard form: $\frac{dl}{dt} + \frac{R}{L}l = \frac{E_0}{L}$. Integrating factor: $\mu(t) = e^{(R/L)t}$.

- 20 Solve the RL circuit equation from above for $I(t)$ given $I(0) = 0$.
 $(e^{(R/L)t}I)' = \frac{E_0}{R}e^{(R/L)t}$, so $e^{(R/L)t}I = \frac{E_0}{R}e^{(R/L)t} + C$, giving $I = \frac{E_0}{R} + Ce^{-(R/L)t}$. Since $I(0) = 0$: $C = -\frac{E_0}{R}$. So $I(t) = \frac{E_0}{R}(1 - e^{-(R/L)t})$.

Mixed review

- 21 Explain why every first-order linear equation $y' + P(x)y = Q(x)$ can, in principle, be solved by the integrating-factor method, regardless of the specific P and Q .
 The integrating factor $\mu(x) = e^{\int P(x) dx}$ is built directly from P , and by construction $\mu' = P\mu$, so $(\mu y)' = \mu y' + \mu' y = \mu(y' + Py) = \mu Q$. Integrating both sides always gives $\mu y = \int \mu Q dx + C$ — the method never fails structurally, though the resulting integral may not have an elementary closed form.
- 22 Solve $y' - 4y = 8x$.
 With $\mu = e^{-4x}$: $(e^{-4x}y)' = 8xe^{-4x}$. By parts, $\int 8xe^{-4x} dx = -2xe^{-4x} - \frac{1}{2}e^{-4x} + C$. So $y = -2x - \frac{1}{2} + Ce^{4x}$.
- 23 Solve $\cos x y' + \sin x y = 1$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. (Divide by $\cos x$ first, and recognize the integrating factor without computing an integral.)
 Standard form: $y' + \tan x y = \sec x$. Here $\mu = e^{\int \tan x dx} = e^{-\ln|\cos x|} = \sec x$ (or note directly that $\mu = \sec x$ makes $(\sec x y)' = \sec x y' + \sec x \tan x y$, matching the equation after multiplying by $\sec x$). Then $(\sec x y)' = \sec^2 x$, so $\sec x y = \tan x + C$, giving $y = \sin x + C \cos x$.

Comparing linear equations with separable equations

- 24 The equation $y' = xy$ is both separable and linear (with $Q(x) = 0$ — a homogeneous linear equation). Solve it using the integrating-factor method and confirm it matches the separable-equation approach.
 Standard form: $y' - xy = 0$, so $\mu = e^{-\int x dx} = e^{-x^2/2}$. Then $(e^{-x^2/2}y)' = 0$, so $e^{-x^2/2}y = C$, giving $y = Ce^{x^2/2}$ — the same family obtained by separating variables directly.
- 25 For which choice(s) of $Q(x)$ does a linear equation $y' + P(x)y = Q(x)$ reduce to a separable equation in general? Explain.
 When $Q(x) = 0$ (the homogeneous case): the equation becomes $y' = -P(x)y$, which separates as $\frac{dy}{y} = -P(x) dx$. For nonzero $Q(x)$, the equation is linear but generally not separable, since y' would depend on both x and y in a way that can't be split into a product of single-variable factors.

Working backward from a solution

- 26 A linear equation $y' + P(x)y = Q(x)$ has general solution $y = x^2 + \frac{C}{x}$. Given that $P(x) = \frac{1}{x}$, find $Q(x)$.
 Differentiate: $y' = 2x - \frac{C}{x^2}$. Then $y' + \frac{1}{x}y = 2x - \frac{C}{x^2} + \frac{1}{x}\left(x^2 + \frac{C}{x}\right) = 2x - \frac{C}{x^2} + x + \frac{C}{x^2} = 3x$. So $Q(x) = 3x$ (consistent with the earlier integrating-factor question above).